Assignment – Week 2

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Preliminary remarks

Please complete **two tasks** for this week. You may choose any two tasks from the assignment, including the core-practice questions. The deadline is **midnight on the 21st of May**. If you want feedback on a particular skill–formal derivation, simulation, or conceptual writing–choose at least one task of that type.

The purpose of these assignments is not only to help you understand the arguments and models covered in the lecture. Much of formal theory is about getting used to notation that can look intimidating at first, and then learning to map it into concrete political examples. A symbol such as R, K, $\kappa(\alpha)$, F, or $\phi(r)$ is useful only if you can say what real-world quantity, institution, or behaviour it is meant to capture. Many of the questions below therefore ask you to move in both directions—from notation to examples, and from examples back to notation.

There is a second habit I would like you to practise. Understanding assumptions by asking when they are violated. Assumptions such as F being continuous, $c''(e) > 0$, or voters observing only a noisy signal are not decorative. They do work in the model. A good answer should therefore be able to say both what the assumption buys us and what would break, change, or become more ambiguous if the assumption failed.

The sections below follow the Week 2 lecture order. Feel free to pick any two tasks, however easy or difficult. For simulation tasks, please include code and the relevant output.

The use of **generative AI is allowed and encouraged as a learning aid**, though it carries risk, like disincentivising proper engagement with the content. A productive workflow is to try the problem yourself first, then use AI to check algebra, debug code, or ask for

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alternative explanations. Please do not submit AI output that you have not understood, edited, and made your own.

Submission instructions: Please email your completed tasks to me by the deadline. Ideally, submit a single integrated document–such as an R Markdown or Jupyter Notebook file–combining code, output, and explanatory text. Essays may also be submitted in Word, LaTeX, or plain Markdown.

1 What Parties Do, and How to Work With Formal Models

Reminder. *The lecture began with the broad question of why parties are useful in representative democracy. It then identified several functions parties can perform, such as allocating scarce political resources, reducing campaign moral hazard, solving commitment problems in multidimensional policy spaces, reducing legislative transaction costs, enabling logrolling, and providing informational shortcuts via party brands.*

Core practice

1. Functions of parties in one table. ***(conceptual)*** Create a table with three columns: party function, one political example, and one model primitive. Fill in at least five rows using the lecture. For example, "seniority-based list positions" might map to a deferred reward R or to a scarce position with cost K.

2. From notation to real-world examples. ***(written)*** Choose **three** objects from the selection story: R, w, c, w_L, w_H, $\underline{\alpha}$, $\bar{\alpha}$, K, η, or v. For each object, answer all three prompts:

(a) What kind of object is it: a choice variable, parameter, type, payoff component, probability, or institutional constraint?

(b) What does it mean intuitively? Give one concrete real-world example that could plausibly be represented by it.

(c) If it became larger, smaller, noisier, or more restrictive, what strategic behaviour would you expect to change?

3. Interpretation. ***(max. 400 words)*** Pick one real party or legislature. Explain which two formal rationales from the lecture best illuminate it. Be specific about the mechanisms rather than simply listing labels.

4. Translating assumptions. ***(written)*** For each assumption below, state (i) what it says mathematically and (ii) what it rules out intuitively:

(a) $c > 0$ and the cost is sunk once paid.

(b) $w_H > w_L$ in the simple selection story.

(c) $\bar{\alpha} > \underline{\alpha}$ in the loyalty–competence extension.

(d) $\eta \in [0, 1)$ affects only high-competence candidates.

2 Sunk Costs and the Loyalty–Competence Trade-off

Reminder. *The main lecture model first used a simple sunk-cost logic: an agent pays a non-recoverable cost c to enter the party track and later chooses between a party reward R and an outside option w. The lecture then added competence, outside options, scarce party slots, and exit risk to show why parties may face a loyalty–competence trade-off.*

Core practice

1. Sunk costs and the margin. ***(written)*** Use the simple two-period sunk-cost model from the lecture. You may take as given that the entry condition is $R - c \geq w$, while the period-2 stay/exit comparison is $R \geq w$.

(a) Explain why c can affect who enters the party track, even though it does not affect the stay/exit comparison once it has already been paid.

(b) Explain the phrase "sunk costs do not matter at the margin" in this model.

(c) Give one real-world political example in which costly early party service might select for people with lower outside options, stronger ideological attachment, or fewer attractive alternatives.

(d) What mistake might an observer make if they saw that long-serving party members rarely exit and concluded that the party had changed their preferences?

2. When does the loyalty–competence trade-off arise? ***(written)*** Take as given the lecture's condition under which the party may prefer to screen out high-competence candidates:

$$(1 - \eta)v\bar{\alpha} < K < v\underline{\alpha}.$$

(a) Explain each part of the condition intuitively. What does $(1-\eta)v\bar{\alpha} < K$ say? What does $K < v\underline{\alpha}$ say?

(b) Why does high competence alone not guarantee that the party wants the high type?

(c) Why is the trade-off sharper when high-competence candidates have better outside options or higher exit risk?

(d) Map the condition to one real-world case: a party, bureaucracy, cabinet, legislature, or candidate-selection setting where loyalty might rationally be prioritised over raw competence.

3. Comparative statics before coding. ***(written)*** Before simulating, predict the direction of the effects. For each change below, say whether it should make screening out high-competence candidates more or less attractive, and why:

(a) Higher exit risk η for high-competence candidates.

(b) Higher outside option $w(\bar{\alpha})$ for high-competence candidates.

(c) Higher scarce-position cost K.

(d) Higher party reward R.

(e) A wider competence gap $\bar{\alpha} - \underline{\alpha}$.

4. First simulation: who enters as sunk costs change? ***(simulation)*** Simulate 10,000 prospective candidates. Half are high competence and half are low competence. High types have outside option $w(\bar{\alpha})$; low types have outside option $w(\underline{\alpha})$; the party reward is R; the sunk cost is c.

Hint. In R, create a grid such as `cgrid <- seq(0, 4, length.out = 500)`; in Python, use `cgrid = np.linspace(0, 4, 500)`. Start with fixed values such as $R = 3$, $w(\underline{\alpha}) = 1$, and $w(\bar{\alpha}) = 2$, then vary one primitive at a time. Store one row per value of c, with columns for the high-type entry rate and low-type entry rate.

(a) Choose baseline values for R, $w(\bar{\alpha})$, and $w(\underline{\alpha})$, and justify them briefly.

(b) Vary c on a grid. For each value of c, compute the share of high and low types who enter.

(c) Plot entry rates by type as a function of c.

(d) Explain the plot using the language of screening and outside options.

(e) Give a real-world example of a party rule that would correspond to raising c.

Extensions

5. Simulating the party's payoff. ***(simulation)*** Extend your entry simulation by assigning values $\bar{\alpha}$ and $\underline{\alpha}$, a productivity multiplier v, a scarce-position cost K, and high-type exit risk η. If a type enters and stays, the party receives $v\alpha - K$; if a high type enters but exits because of the exit shock, the party receives $-K$.

Hint. Write one function that takes $(c, K, \eta, R, w(\bar{\alpha}), w(\underline{\alpha}))$ and returns expected party payoff. Then call the same function on different grids. This is easier to debug than writing separate code for each comparative static. Use a fixed seed if you draw candidate types randomly.

(a) Vary c, K, η, $w(\bar{\alpha})$, and $w(\underline{\alpha})$ one at a time. For each variation, plot the party's expected payoff.

(b) Identify parameter regions in which the payoff-maximising c mainly attracts low types.

(c) Identify parameter regions in which the party wants both types, and explain why.

(d) Compare your simulated patterns to the condition $(1 - \eta)v\bar{\alpha} < K < v\underline{\alpha}$.

(e) Map one parameter change to a real institutional reform, such as raising the cost of nomination, increasing salaries, severing outside ties, or using more flexible appointments.

6. Institutional design. ***(essay, max. 1,000 words)*** The lecture listed several tools parties use to screen candidates, such as unpaid party work, ideological signalling, party schools, seniority-based promotion, trial appointments, internal endorsements, and severing outside ties. Choose two tools. For each, explain which primitive it most directly affects–c, R, K, $w(\alpha)$, η, or the informativeness of effort–and discuss one way the tool could backfire. Use at least one concrete real-world application.

3 Logrolling, Preference Intensity, and Party Discipline

Reminder. *Logrolling means trading support across issues, i.e. one actor supports a policy she values less in exchange for another actor's support on a policy she values more. The key modelling idea is not only that actors disagree about positions, but also that they may care with different intensity about different issues. Party discipline can make such cross-issue bargains credible.*

Core practice

1. Logrolling in one real bill. ***(written)*** Choose one real bill or legislative package and rationalise it through the logrolling lens. Good starting points include the US Inflation Reduction Act, the Affordable Care Act, US farm bills, EU budget packages, German coalition agreements, or omnibus spending bills. In one paragraph, identify the actors, the issues bundled together, what each actor gives up, and what each actor receives. Then translate the example into a two-dimensional policy space with coordinates (x, y).

2. From issue positions to preference intensity. ***(written)*** Suppose actor i has utility

$$U_i(x, y) = -[\lambda_i(x - x_i)^2 + (1 - \lambda_i)(y - y_i)^2],$$

where $\lambda_i \in (0, 1)$ measures the intensity with which actor i cares about issue x relative to issue y.

(a) Explain what λ_i means politically.

(b) What does $\lambda_i > 1/2$ imply? What does $\lambda_i < 1/2$ imply?

(c) Give a real-world example of two parties that may have similar positions on one issue but different preference intensities across issues.

(d) Explain why differences in preference intensity can create room for a bargain even when both sides dislike parts of the final package.

3. Simulating a small logroll. ***(simulation)*** Create two actors, A and B, with ideal points of your choice in $[0, 1]^2$. Choose preference-intensity parameters λ_A and λ_B such that A cares more about issue x and B cares more about issue y.

Hint. Use a grid such as $x, y \in \{0, 0.01, 0.02, \ldots, 1\}$. In R, `expand.grid()` is useful; in Python, use `np.meshgrid`. Store the default utility for each actor, then create a Boolean variable for whether each grid point improves on the default.

(a) Define a default policy bundle.

(b) Evaluate both actors' utilities on a grid over $[0, 1]^2$.

(c) Shade the set of bundles that both actors prefer to the default.

(d) Choose one bundle in that set and explain the political exchange it represents.

(e) Map your setup to a real-world bargaining example.

4. Why discipline matters. ***(written, max. 500 words)*** Suppose a logroll must be implemented through two separate votes rather than a single package. Explain why the bargain can unravel after the first vote. Then identify one party institution–whipping, agenda control, list placement, committee assignment, coalition agreement, or repeated career incentives–that could make the bargain credible. Apply the argument to a real legislature.

Extensions

5. Preference intensity and the size of the bargaining region. ***(simulation)*** Hold ideal points fixed and vary λ_A and λ_B on a grid. For each pair of intensity parameters, compute the area of the set of policy bundles that both actors prefer to the default.

(a) Plot the bargaining-region area as a function of (λ_A, λ_B).

(b) When is the bargaining region largest?

(c) When is it empty or nearly empty?

(d) Explain the result in terms of preference intensity, not just spatial distance.

(e) Give a real-world interpretation of one case with a large bargaining region and one case with a small bargaining region.

6. Efficient logrolling or strategic gridlock? ***(essay, max. 1,000 words)*** Levy (2004) treats parties as commitment devices that enable efficient cross-issue deals. Lee (2022) argues that bundling can also create gridlock when actors delay good policies to gain leverage on divisive ones. Are the two claims compatible? Identify the conditions under which each story is more plausible, and use one real-world case to illustrate your answer.

4 Transaction Costs and Party Brands

Reminder. *The lecture then gave two further rationales for parties. A Coasean view says that parties reduce the transaction costs of repeated legislative bargaining. An informational view says that party labels help voters infer candidate positions when individual information is costly or noisy. Snyder and Ting's argument is that party labels are valuable only when they are informative: a label helps voters if it credibly restricts the kinds of candidates who can run under it.*

Core practice

1. What is the "transaction" in the Coasean story? ***(written)*** Boix's analogy is to Coase's theory of the firm: when market contracting is costly, hierarchy can sometimes organise activity more efficiently. In legislatures, the relevant "transactions" are bargains over votes, information, agenda time, committee work, coalition maintenance, and policy concessions.

(a) Choose one legislature or party system.

(b) Identify three transactions legislators repeatedly need to make.

(c) For each transaction, name the cost involved: search costs, bargaining costs, monitoring costs, enforcement costs, agenda costs, or information costs.

(d) Explain what the party organisation does to reduce each cost.

(e) Give one example where the party might increase costs instead, for example by enforcing discipline, suppressing internal information, or blocking cross-party bargains.

2. What can parties do that contracts cannot? ***(short essay, max. 700 words)*** The Coasean analogy is useful, but political parties are not just private firms. They control nominations, career ladders, committee assignments, access to donors and activists, shared reputations, and sometimes voters' partisan identities. Choose two of these tools. For each,

explain why a simple external contracting agency would struggle to replicate it. Then apply the argument to a real party or legislature.

3. Coasean predictions across institutions. ***(written)*** Compare a parliamentary system with strong party discipline to a presidential system with weaker formal party control. Use the Coasean view to answer:

(a) Where should parties be more valuable as transaction-cost reducers?

(b) Which costs should be especially high without parties?

(c) Which institutional features substitute for parties?

(d) Give one observable implication: what would you expect to see in committee work, agenda control, coalition bargaining, or legislative productivity?

4. Building the Snyder–Ting intuition. ***(written)*** Consider a simplified candidate-information problem. Without party labels, individual candidates' true positions are placed on the policy interval according to

$$\theta_j \sim \text{Uniform}[0, 1].$$

Voters do not observe θ_j directly. Instead, they observe a noisy signal

$$z_j = \theta_j + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \tau^2).$$

A party label P tells voters that candidates running under that label come from a narrower interval around the party brand:

$$\theta_j \mid P \sim \text{Uniform}\left[\mu_P - h_P, \mu_P + h_P\right] \cap [0, 1].$$

(a) Explain what is random in this model and what is common knowledge.

(b) In words, what does τ measure? What does h_P measure?

(c) Why does a smaller h_P make the party label more informative?

(d) Why would the label become uninformative if any candidate could freely adopt it?

(e) Map h_P to a real-world party feature: ideological discipline, nomination control, factional diversity, candidate autonomy, or brand dilution.

5. Voter mistakes without parties. ***(numerical)*** Assume one voter with ideal point $x_i = 0.5$ chooses between two individual candidates with true positions $\theta_1 = 0.4$ and $\theta_2 = 0.7$. The voter observes $z_1 = \theta_1 + \varepsilon_1$ and $z_2 = \theta_2 + \varepsilon_2$ and votes for the candidate whose signal is closer to x_i.

(a) Without computing an exact probability, explain what it means for the voter to make a mistake here.

(b) If τ rises, should mistakes become more or less likely? Explain.

(c) If the voter also observes that candidate 1 belongs to a party with $\mu_A = 0.4$ and small h_A, how should that change the voter's estimate of θ_1?

6. Brand dilution. ***(written, max. 600 words)*** Lupu (2016) argues that Latin American party brands were diluted by ideological flip-flops. In this model, is brand dilution best represented by a change in μ_P, h_P, τ, or some combination? Explain your answer and state one empirical implication for electoral volatility.

Extensions

7. Simulating Coasean transaction costs. ***(simulation)*** Build a simple simulation of legislative bargaining. Suppose N legislators must form majority coalitions across B bills. Without parties, each bill requires a new coalition search. With parties, legislators are grouped into party blocs that reduce search costs but may impose discipline costs on dissenters.

Hint. Start with fixed values, for example $N = 101$, $B = 100$, two parties, and a per-negotiation search cost of 1. Then vary one parameter at a time. Store one row per parameter combination with columns for no-party bargaining costs, party bargaining costs, discipline costs, and total costs.

(a) Choose values for N, B, the number of parties, and a per-negotiation search cost.

(b) Simulate total bargaining costs with no parties and with parties.

(c) Add a discipline cost paid when a legislator votes against her ideal position because of party pressure.

(d) Vary party size and party discipline. When do parties reduce total costs? When do they merely shift costs from bargaining to discipline?

(e) Map one simulation parameter to a real institutional feature.

8. Party labels and vote-choice error. ***(simulation)*** Simulate an election with voters uniformly distributed on $[0, 1]$. Compare two environments.

(a) *No party labels:* each candidate has a true position θ_j, but voters observe only noisy signals $z_j = \theta_j + \varepsilon_j$.

(b) *Party labels:* candidates belong to parties with brand means μ_P and brand half-widths h_P, and voters combine the candidate signal with the party interval in a transparent way of your choice. State your rule clearly.

For two candidates located at 0.35 and 0.65, simulate the share of voters who choose the candidate farther from their true ideal point. Vary τ and h_P. Plot mistake rates. What does the simulation say about when party brands are most valuable?

9. Connecting transaction costs and brands. ***(essay, max. 900 words)*** Party brands are usually discussed as voter-facing informational shortcuts. Could they also reduce transaction costs inside legislatures? Write a short essay that uses one concrete case to argue either yes or no.

5 The Downsian Model and Its Scope Conditions

Reminder. *The second half of the lecture introduced the Downsian benchmark: two vote-maximising parties choose positions $x_A, x_B \in [0, 1]$; voters choose the closer party; voter ideal points are distributed according to F; and, with a continuous strictly increasing F, both parties converge to the median m where $F(m) = 1/2$.*

Core practice

1. Nash equilibrium intuitively. ***(written)*** Define a pure-strategy Nash equilibrium in the two-party Downsian game without using algebra. Then explain why the word "simultaneously" matters in the setup.

2. Computing vote shares by hand. ***(numerical)*** Assume voters are uniformly distributed on $[0, 1]$. For each pair below, compute the indifferent voter $\hat{x} = (x_A + x_B)/2$, each party's vote share, and one profitable deviation for A if one exists:

(i) $x_A = 0.3$, $x_B = 0.6$;

(ii) $x_A = 0.4$, $x_B = 0.55$;

(iii) $x_A = 0.45$, $x_B = 0.55$;

(iv) $x_A = 0.50$, $x_B = 0.55$.

Where is A's best response when $x_B = 0.55$?

3. Finding medians. ***(numerical/conceptual)*** Identify the median m for each distribution:

(a) $F = \text{Uniform}[0, 1]$;

(b) $F = \text{Beta}(2, 2)$;

(c) $F = \text{Beta}(2, 5)$, using $m \approx 0.264$ if helpful;

(d) a 50/50 mixture of point masses at 0.2 and 0.8;

(e) a symmetric bimodal mixture centred at 0.3 and 0.7.

For the point-mass case, explain why the median is not unique.

4. Scope conditions. ***(conceptual)*** The lecture grouped departures from the Downsian model into contextual, demand-side, and supply-side assumptions. Pick one assumption from each category. For each, state the assumption, give one real-world setting where it may fail, and say what kind of party behaviour you would expect instead of simple median convergence.

5. Valence intuition. ***(written + numerical)*** Voter utility is now $U_i(x_j) = -|x_i - x_j| + v_j$. Let $x_A < x_B$ and $\Delta v = v_A - v_B \geq 0$.

(a) Derive the indifferent voter $\hat{x} = \frac{x_A + x_B}{2} + \frac{\Delta v}{2}$.

(b) Plug in $x_A = 0.3$, $x_B = 0.7$, and $\Delta v = 0.1$. Compute each party's vote share.

(c) Explain why a high-valence party can sometimes stand further from the median and still win.

Extensions

6. Visualising convergence and breakdown. ***(simulation)*** Draw voters from three distributions: Uniform$[0, 1]$, Beta$(2, 5)$, and a symmetric bimodal mixture centred at 0.3 and 0.7. For each, compute the empirical median and use a grid search to find each party's best response to a range of opponent positions. Plot the best-response curves or an iterative best-response trajectory. What changes across distributions, and what does not?

Hint. Use a grid for party positions, such as `xgrid = np.linspace(0, 1, 501)` in Python or `xgrid <- seq(0, 1, length.out = 501)` in R. For each opponent position, compute vote share at each possible response and choose the response that gives the highest vote share. Start with the uniform case before adding the skewed and bimodal distributions.

7. Persistent divergence. ***(essay, max. 1,000 words)*** Some two-party systems appear to converge toward the median; others, such as Australia in Mussel and Schlechta (2024), show persistent divergence despite institutional features that seem favourable to Downsian logic. Choose two scope conditions and explain whether persistent divergence is best understood as a failure of the model or as evidence that those conditions do not hold.

Optional Challenges

The questions in this section draw more directly on appendix material. They are optional stretch tasks. You may submit one of them as one of your two tasks, but you do not need to use this section to complete the assignment.

A1. Separating signals of competence. ***(formal)*** In the appendix signalling game, a high type has effort cost $\kappa(\bar{\alpha})$ and a low type has effort cost $\kappa(\underline{\alpha})$, with $\kappa(\bar{\alpha}) < \kappa(\underline{\alpha})$. The party promotes after $e = 1$ and does not promote after $e = 0$. Derive the agent-side condition for separation and interpret it intuitively.

A2. Politicians' pay. ***(essay, max. 900 words)*** Use the entry condition $W + m_i \geq w(a_i) + k$ to explain why raising politicians' salaries may improve average competence but have ambiguous effects on public-service motivation. What kind of empirical evidence would help distinguish the two channels?

A3. Discipline, unravelling, and distributive bargains. ***(simulation)*** Simulate a two-bill logroll in which each actor receives benefit b_i from her preferred bill and pays cost

d_i from supporting the other actor's bill. Let party discipline impose a defection penalty $P \geq 0$.

(a) Draw b_i and d_i from distributions of your choice and justify those choices.

(b) For each simulated bargain, classify whether it is jointly valuable, whether it unravels without discipline, and whether it is enforceable for a given P.

(c) Vary P and plot the share of bargains that become enforceable.

(d) Separate cases in which discipline enforces mutually beneficial bargains from cases in which it mainly enforces distributive transfers.

(e) Apply the result to one real-world party or coalition.

A4. How many party labels? ***(simulation)*** This is a deliberately simplified way to think about the number of parties. Suppose voters are distributed on $[0, 1]$ and candidate positions are represented by J party labels located at equally spaced brand means $\mu_1, \ldots, \mu_J$. A voter chooses the party whose perceived position is closest to her ideal point. Perceived positions are noisy:

$$\tilde{\mu}_j = \mu_j + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \tau^2).$$

Define representational loss as the average distance between each voter's ideal point and the true position of the party she chooses.

(a) Compare $J \in \{2, 3, 4, 5, 6\}$ when voters are uniformly distributed.

(b) Repeat when voters are drawn from a bimodal distribution centred at 0.3 and 0.7.

(c) Repeat for several values of τ.

(d) Which J minimises representational loss in each case? How does the answer depend on the distribution of ideal points and the information-noise parameter τ?

(e) What important political forces are missing from this simplified exercise? Name at least two.

A5. Convergence and uniqueness. ***(formal)*** Prove the median voter theorem for the baseline model. Your answer should show: (i) why $x_A = x_B = m$ is a Nash equilibrium, and (ii) why any profile away from the median gives at least one party a profitable deviation. Be explicit about where continuity and strict monotonicity of F enter the argument.

A6. Valence and one-sided convergence. ***(formal + essay, max. 900 words)*** Suppose A has a fixed valence advantage over B and may also be policy-motivated. How far from the median can A locate while still winning a majority, as a function of Δv? Then connect the result to one empirical example in which perceived competence, credibility, or leadership quality changed the amount of policy convergence we observe.

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